

*MA.7.AR.4.5***Benchmark****MA.7.AR.4.5 Solve real-world problems involving proportional relationships.**

Example: Gordy is taking a trip from Tallahassee, FL to Portland, Maine which is about 1,407 miles. On average his SUV gets 23.1 miles per gallon on the highway and his gas tanks holds 17.5 gallons. If Gordy starts with a full tank of gas, how many times will he be required to fill the gas tank?

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.2.2
- MA.7.AR.3.3
- MA.7.GR.1.5
- MA.7.GR.2.2, MA.7.GR.2.3
- MA.7.DP.1.4
- MA.8.AR.3.4, MA.8.AR.3.5

Terms from the K-12 Glossary

- Proportional Relationships
- Rate
- Unit Rates

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.5
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.AR.3.1, MA.7.AR.3.2

Next Benchmarks

- MA.912.AR.2.4, MA.912.AR.2.5
- MA.912.F.1.5

Purpose and Instructional Strategies

In previous courses, students solved mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system. In grade 7 accelerated, students solve real-world problems involving proportional relationships. Students will solve real-world problems involving linear relationships. Students also interpret the slope and y-intercept of a two-variable linear equation within a real-world context when given a written description, a table, a graph or an equation. In Algebra 1, students will solve mathematical and real-world problems that are modeled by linear functions.

- This benchmark is a culmination of the work students have been doing throughout MA.7.AR.4.
- Instruction for this benchmark includes opportunities to compare two different proportional relationships to each other.
- Allow various methods for solving, encouraging discussion and analysis of efficient and effective solutions (*MTR.4.1*).
- In grade 7, it is not the expectation that students define independent and dependent variables. Students may have experience with these terms from other content areas.

Common Misconceptions or Errors

- Students may confuse the dependent and independent variables when graphing. To address this conception, instruction includes the understanding that the independent variable depends on the given context. Additionally, independent variables are not always on the x -axis and the dependent variables are not always on the y -axis.
 - For example, if one has a proportional relationship between feet and meters, they can graph feet either on the x -axis or the y -axis. Which one that is dependent depends on the context. For instance, if one is given feet and converting to meters, then feet would be independent and meters would be dependent.

Strategies to Support Tiered Instruction

- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions.
 - What do you know from the problem?
 - What is the problem asking you to find?
 - What are the two quantities in this problem?
 - How are the quantities related to each other?
 - Which quantity do you want to consider as the independent variable?
 - Which quantity do you want to consider as the dependent variable?
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Instruction includes the understanding that the independent variable depends on the given context. Additionally, independent variables are not always the x -axis and the dependent variable are not always the y -axis.
 - For example, if one has a proportional relationship between feet and meters, they can graph feet either on the x -axis or the y -axis. Which one is dependent depends on the context. For instance, if one is given feet and converting to meters, then feet would be independent, and meters would be dependent.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Cassie is making shortbread cookies using the ingredients below.

10 tablespoons of butter	$1\frac{1}{2}$ cups flour
$1\frac{1}{2}$ cups powdered sugar	$\frac{1}{2}$ teaspoon salt
$\frac{1}{2}$ teaspoon vanilla extract	

Part A. This recipe makes 16 cookies, but Cassie needs 5 dozen. How much of each ingredient will she need to make the 5 dozen cookies she needs?

Part B. Once Harrison tasted Cassie's shortbread cookies, he ordered 7 dozen for a birthday party. If Cassie originally started with 4 cups of flour, 2 cups of powdered sugar and 16 tablespoons of butter, how much more (if any) will she need of each ingredient to complete Harrison's order?

Part C. After the party, Cameron shared his recipe which calls for 2 cups of flour and $1\frac{3}{4}$ cup of powdered sugar. Since adding powdered sugar to cookies should make them sweeter, Cameron claims his larger ratio of powdered sugar to flour will produce sweeter cookies. Is this statement correct?

Instructional Items

Instructional Item 1

A couple is taking a horse and carriage ride through Central Park in New York City. After 8 minutes, they had traveled $\frac{1}{2}$ mile.

Part A. Create a graph to represent the proportional relationship between miles traveled and the number of minutes they are on the carriage.

Part B. Use this graph to determine how long it will take to complete the 2.5-mile ride around the park.

Instructional Item 2

Trey is learning new music for his band class. He has 45 measures of music to learn. Last week, he learned 14 measures in 21 minutes on Monday, 9 measures in 13 minutes and 30 seconds on Tuesday, 11 measures in 16 minutes and 30 seconds on Wednesday, 3 measures in 4 minutes and 30 seconds on Thursday, and 8 measures in 12 minutes on Friday.

Determine if there is a proportional relationship in his music practice during this week.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*